MITS 6700G - Assignment 2

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Complex Networks - MITS 6700G

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1.a) since the calulation for ~~page~~ rank is, $PR(node) = \sum \frac{PR(\text{Incoming})}{\text{outdegree(of that node)}}$

so,

nodes which don't have any incoming node will have the page rank of 0.

Nodes with page rank are:

Node, 6, 11, 12 & 17

because they don't have any incoming so their page rank will be 0.

1.b ~~Iteration 1.~~

~~Node: page rank(Initial)(I1)~~ ~~I2~~ ~~I3~~

Initial page rank $= 1/18$ for all 18 nodes.

$$PR(1) = \frac{PR(4)}{\text{outdegree}(4)}$$

Node	Incoming Node	PR(Incoming Node)	PR(Node)
1	4	$\frac{PR(4)}{\text{outdegree of } 4} = \frac{1/18}{3} = \frac{1}{54}$	$1/54$
2.	6, self	$\frac{PR(6)}{\text{out degree of } 6} = \frac{1/18}{2} = \frac{1}{36}$	$1/36 + 1/18 = 1/12$
3	4 & 9	$\frac{PR(4)}{\text{out degree }(4)} + \frac{PR(9)}{\text{out degree }(9)}$	$\frac{1/18}{3} + \frac{1/18}{3} = \frac{2}{54} \frac{1}{27}$
4.	9	$\frac{PR(9)}{\text{out degree}(9)} = \frac{1/18}{3}$	$1/54$

Node	Incoming node	PR(Incoming)	PR(Node)
5	4	$\frac{PR(4)}{outdegree(4)} = \frac{1/18}{3}$	$1/54$
6	0	0	0
7	6 11	$\frac{PR(6)}{outdegree} = 1/18/2 = 1/36$ $\frac{PR(11)}{outdegree(11)} = 1/18$	$1/36 + 1/18 = 3/36$ $= 1/12.$
8	1, 7, 3	$\frac{PR(1)}{outdegree(1)} = 1/18$ $\frac{PR(7)}{outdegree(7)} = 1/18$ $\frac{PR(3)}{outdegree(3)} = 1/18$	$1/18 + 1/18 + 1/18$ $= 3/18 = 1/6.$
9	14	$\frac{PR(14)}{outdegree(14)} = \frac{1/18}{1}$	$= 1/18$
10	5, ~~0~~ 16 & slef	$\frac{PR(5)}{outdegree(5)} = 1/18$ $\frac{PR(16)}{outdegree(16)} = 1/18$	$1/18 + 1/18 = 2/18$ $= 1/9 + 1/18$ $= 1/6.$
11	0	0	0
12	0	0	0
13	8, 12, 18	$\frac{PR(8)}{outdegree(8)} = \frac{1/18}{1} = 1/18$ $\frac{PR(12)}{outdegree(12)} = 1/18$ $\frac{PR(18)}{outdegree(18)} = 1/18$	$1/18 + 1/18 + 1/18$ $3/18 = 1/6$

Node	Inioming Node	PR(Inioming)	PR(Node)
14	13	$\frac{PR(13)}{outdegree(13)} = \frac{1/18}{1} = \frac{1}{18}$	$\frac{1}{18}+\frac{1}{54}=$ [illegible]
	15	$\frac{PR(15)}{outdegree(15)} = \frac{1/18}{3} = \frac{1}{54}$	$\frac{4}{54}$ [illegible]
15	9	$\frac{PR(9)}{outdegree(9)} = \frac{1}{54}$	$\frac{1}{54}$
16	15	$\frac{PR(15)}{outdegree(15)} = \frac{1}{54}$	$\frac{1}{54}+\frac{1}{18}$
	17	$\frac{PR(17)}{outdegree(17)} = \frac{1}{18}$	$\frac{2}{27}$
17	0	0	0
18	15	$\frac{PR(15)}{outdegree(15)} = \frac{1}{54}$	$\frac{1}{54}$.

Iteration 2 :->

Node	Inioming Node	PR(Inioming)	PR(Node)
1	4	$\frac{PR(4)}{outdegree(4)} = \frac{1/54}{3}$	$\frac{1}{162}$
2	6 & self	$\frac{PR(6)}{outdegree(6)} = 0$	$0+\frac{1}{2}=\frac{1}{2}$
3	4	$\frac{PR(4)}{outdegree(4)} = \frac{1/54}{3}$	$\frac{1}{162}+\frac{1}{54}$
	9	$\frac{PR(9)}{outdegree(9)} = \frac{1/18}{3}$	$\frac{4}{162}=\frac{2}{81}$

Node	Incoming Node	PR(Incoming)	PR(Node)
4	9	$\frac{PR(9)}{outdegree(9)} = \frac{1/18}{3}$	$1/54$
5	4	$\frac{PR(4)}{out\ degree(4)} = \frac{1/54}{3}$	$1/162$
6	0	0	0
7	6 11	$\frac{PR(6)}{outdegree(6)}, \frac{PR(11)}{outdegree(11)}$	0
8	1, 7, 3	$\frac{PR(1)}{outdegree(1)} = 1/54$ $\frac{PR(7)}{outdegree(7)} = 1/12$ $\frac{PR(3)}{outdegree(\cancel{2}3)} = 1/27$	$\frac{2+9+4}{108} = \frac{15}{108}$ $5/36$
9	14	$\frac{PR(14)}{outdegree(14)} = \frac{2/27}{1}$	$2/27$
10	5, 16 & self	$\frac{PR(5)}{outdegree(5)} = 1/54$ $\frac{PR(16)}{outdegree(16)} = 2/27$	$1/54 + 2/27 = 5/54$ $5/54 + 1/6 = 14/54$ $= 7/27$
11	0	0	0
12	0	0	0
13	8, 12, 18	$\frac{PR(8)}{outdegree(8)} = 1/6$ $\frac{PR(12)}{outdegree(12)} = 0$ $\frac{PR(18)}{outdegree(18)} = 1/54$	$1/6 + 1/54 = \frac{9+1}{54}$ $= \frac{10}{54} = 5/27$

Node	Incoming Node	PR(Incoming)	PR(Node)
14	13	$\frac{PR(13)}{outdegree(13)} = 1/\cancel{18}6$	$1/\cancel{18}6 + 1/162$
	15	$\frac{PR(15)}{outdegree(15)} = \frac{1/54}{3} = 1/162$	$\frac{\cancel{27}\,\cancel{9}+1}{162} = \cancel{0.28}$
15	9	$\frac{PR(9)}{outdegree(9)} = \frac{1/18}{3}$	$1/54$
16	15	$\frac{PR(15)}{outdegree(15)} = 1/162$	$1/162$
	17	$\frac{PR(17)}{outdegree(17)} = 0$	
17	0	0	0
18	15	$\frac{PR(15)}{outdegree(15)} = 1/162$	$1/162$

Iteration 3 -

Node	Incoming Node	PR(Incoming)	PR(Node)
1	4	$\frac{PR(4)}{outdegree(4)} = \frac{1/54}{3}$	$1/162$
2	6 & self	$0 + 1/12$	$1/12$
3	4	$\frac{PR(4)}{outdegree(4)} = 1/162$	$1/162 + 2/81$
	9	$\frac{PR(9)}{outdegree(9)} = \frac{2/27}{3} = 2/81$	$1/162 + 4/162$ $5/162$

Node	Incoming Node	PR(Incoming)	PR(Node)
4	$\cancel{\frac{PR(9)}{\text{outdegree}}}$ 9	$\frac{PR(9)}{\text{outdegree}(9)} = 2/81$	$2/81$
5	4	$\frac{PR(4)}{\text{outdegree}(4)} = 1/162$	$1/162$
6	0	0	0
7	6	$\frac{PR(6)}{\text{outdegree}(6)} = 0$	0
	11	$\frac{PR(11)}{\text{outdegree}(11)} = 0$	
8	1, 7, 3	$\frac{PR(1)}{\text{outdegree}(1)} = 1/162$	$1/162 + 4/162$
		$\frac{PR(7)}{\text{outdegree}(7)} = 0$	$5/162$
		$\frac{PR(3)}{\text{outdegree}(3)} = 2/81$	
9	14	$\frac{PR(14)}{\text{outdegree}(14)} = \frac{28}{162}$	$\frac{14}{81}$
10	5, 16, & self	$\frac{PR(5)}{\text{outdegree}(5)} = 1/162$	$2/162 = 1/81 + 7/27 = 22/81$
		$\frac{PR(16)}{\text{outdegree}} = 1/162$	
11	0	0	0
12	0	0	0

$\frac{14}{81}$

Node	Incoming Node	PR(Incoming)	PR(Node)
13	8, ~~[illegible]~~, 12 & 18	$\frac{PR(8)}{outdegree(8)}$ $\frac{PR(12)}{outdegree(12)}$ $\frac{PR(18)}{outdegree(18)} = 1/162$	$5/36$ ~~+0~~ $+ 1/162$ $\frac{45+2}{324}$ $= \frac{47}{324}$
14	13	$\frac{PR(13)}{outdegree(13)} = 5/27$	~~$5/27 + 1/243$~~ ~~$= 44/243$~~ ~~$= 53/243$~~
	15	$\frac{PR(15)}{outdegree(15)} = $ ~~[illegible]~~ $= 1/162$	$\frac{5}{27} + 1/162$ $= 31/162$
15	9	$\frac{PR(9)}{outdegree(9)} = 2/81$	$2/81$
16	15	$\frac{PR(15)}{outdegree(15)} = 1/162$	$1/162$
	17	$\frac{PR(17)}{outdegree(17)} = 0$	
17	0	0	0
18	15	$\frac{PR(15)}{outdegree(15)}$ -	$1/162$

Node	Iteration 1	Iteration 2	Iteration 3
1	1/54	1/162	1/162
2	1/12	1/12	1/12
3	1/27	2/81	5/162
4	1/54	1/54	2/81
5	~~1/162~~ 1/54	1/162	1/162
6	0	0	0
7	1/12	0	0
8	1/6	~~2/27~~ 5/36	5/162
9	1/18	2/27	14/81
10	1/6	~~5/54~~ 7/27	~~4/81~~ 22/81
11	0	0	0
12	0	0	0
13	1/6	5/27	47/324
14	2/27	14/81	31/162
15	1/54	1/54	2/81
16	2/27	1/162	1/162
17	0	0	0
18	1/54	1/162	1/162

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After iteration 2: 6, 7, 11, 12, and 17 have page rank value: 0.

After Iteration 3: 6, 7, 11, 12 & 17 have page rank value of 0.

(c) Nodes with non-zero-page rank values are: node: 1, 2, 3, 4, 5, 8, 9, 10, 13, 14, 15, 16 & 18

On summarizing, nodes 6, 7, 11, 12 & 17 will never have any incoming edges that carries a page rank value (non-zero) & due to this these node will always remain 0.

(d) Nodes 5 and 16 if we remove the links either or both from 5 and 16 it will increase their equilibrium page rank value. Because both 5 & 16 goes to a "sink" node with no outgoing link i.e node 10.

(e) The minimum number of edges that needs to be added is 5.
2 edges for 2 sink nodes 2 & 10.
followed by 3 edges for dangling nodes they are 6, 11 and 17.

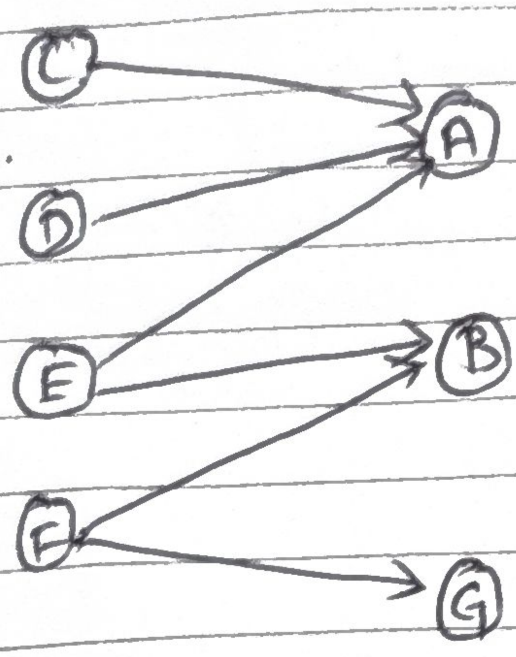
(7) Using scaling factor of $s = 0.9$, which means that there is a $1-s$ i.e, 0.1 ~~chance that~~ or 10% chance that any node can randomly create edge with any of other 17 node. So, we add $\frac{1-s}{n}$ which is:

$$\frac{1-0.9}{18} = \frac{0.1}{18}$$ to the pag rank

for all the nodes. Hence, there will be no any node with page rank 0.

Node 10 will now have the highest page rank value because it keeps all the weight from it's neighboring node 5 and 16 which will increase it rank value coz it doesn't have any outgoing link.

Q.No.2



(a) Here,

we have,

Hubs = C, D, E, F

Authorities = A, B, G

~~Giving the value of 1 as~~

Iteration 1

Initial Authority for all node of 1.

$a(A) = h(C) + h(D) + h(E) = 3$

$a(B) = h(E) + h(F) = 2$

$a(G) = h(F) = 1.$

$h(C) = a(A) = 3$

$h(D) = a(A) = 3$

$h(E) = a(A) + a(B) = 5$

$h(F) = a(B) + a(G) = 3$

~~Normal~~

Hub total $= h(C) + h(D) + h(E) + h(F)$
$= 3 + 3 + 5 + 3 = \underline{14}$.

Authority total $= a(A) + a(B) + a(G)$
$= 3 + 2 + 1$
$= 6$

Normalizing using L1:

Authorities

$a(A) = 3/6 = 1/2$
$a(B) = 2/6 = 1/3$
$a(G) = 1/6$

Hubs

$h(C) = 3/14$
$h(D) = 3/14$
$h(E) = 5/14$
$h(F) = 3/14$

__Iteration 2__

$a(A) = h(C) + h(D) + h(E) = 3+3+5 = \underline{11}/14$
$a(B) = h(E) + h(F) = 5 + 3 = 8/14$
$a(G) = h(F) = 3/14$.

$\frac{11}{14} + \frac{8}{14} + \frac{3}{14}$
$\frac{22}{14}$ $\frac{22}{14}$
$1/2$,

$h(C) = a(A) = \underline{11}$
$h(D) = a(A) = \underline{11}$
$h(E) = a(A) + a(B) = 11 + 8 = \underline{19}$
$h(F) = a(B) + a(G) = 8 + 3 = \underline{11}$.

Hub total $= \underline{11} + \underline{11} + \underline{11} + \underline{19} = 52$
Authorities total $= \underline{11} + 8 + 3 = 22$.

Normalizing using ~~L2~~ L1:

<u>Authorities</u>

$a(A) = 11/22 = 1/2$

$a(B) = 8/22 = 4/11$

$a(G) = 3/22$

<u>Hubs</u>

$h(C) = 11/52$

$h(D) = 11/52$

$h(E) = 19/52$

$h(F) = 11/52$

	~~Iter~~ Iteration	1	Iteration	2
	(Before)	(After)	(Before)	(After)
a(A)	3	1/2	11	1/2
a(B)	2	1/3	8	4/11
a(G)	1	1/6	3	3/22
h(C)	3	3/14	11	11/52
h(D)	3	3/14	11	11/52
h(E)	5	5/14	19	19/52
h(F)	3	3/14	11	11/52

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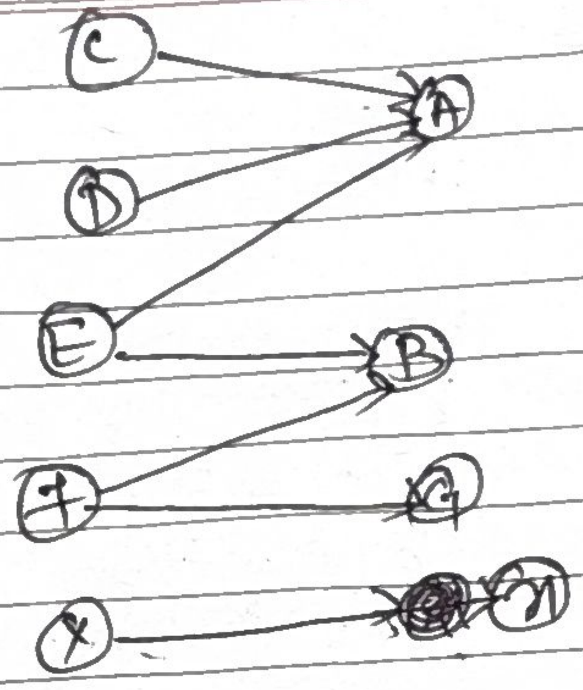
2. (b) (i)



Iteration 1

$a(H) = h(X) = 1$ $\quad$ $h(C) = 3$

$a(A) = 3$ $\quad$ $h(d) = 3$

$a(B) = 2$ $\quad$ $h(e) = 5$

$a(G) = 1$ $\quad$ $h(f) = 3$

$a(H) = 1$ $\quad$ $h(Y) = 1.$

authority total $= 3+2+1+1 = 7$ $\{a(A)+a(B)+a(G)+a(H)\}$

hub total $= h(c)+h(d)+h(e)+h(f)+h(Y) = 3+3+5+3+1$
$= 15$

Authority

so, $a(X) = 1/7$

$a(A) = 3/7$

$a(B) = 2/7$

$a(G) = 1/7$

$a(H) = 1/7$

Hubs

$h(c) = 3/15 = 1/5$

$h(d) = 3/15 = 1/5$

$h(e) = 5/15 = 1/3$

$h(f) = 3/15 = 1/5$

$h(Y) = 1/15$

Iteration 2

$a(H) = 1$ $\quad$ $h(C) = 11$

$a(A) = 11$ $\quad$ $h(D) = 11$

$a(B) = 8$ $\quad$ $h(E) = 19$

$a(G) = 3$ $\quad$ $h(F) = 11$

$h(Y) = 1$

authority total = 1+11+8+3 = 23

hub total = ~~53~~ 11+11+19+11+1 = 53.

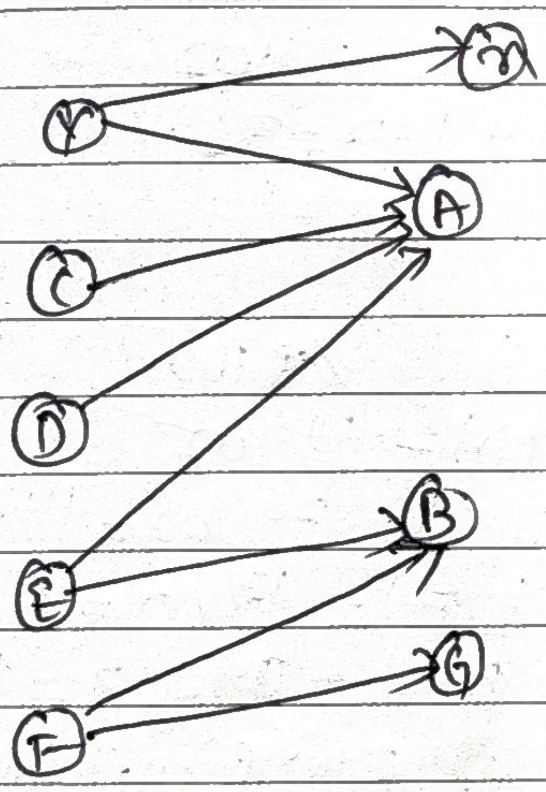
Normalized value:

Authority	Hub
a(A) = 11/23	h(C) = 11/53
a(B) = 8/23	h(D) = 11/53
a(G) = 3/23	h(E) = 19/53
a(n) = 1/23	h(F) = 11/53
	h(Y) = ~~1/53~~ 1/53

Hence, we have,

Iteration 1	(Before)	(After)	Iteration 2 (Before)	(After)
a(n) =	1	1/7	1	1/23
h(Y) =	1	1/15	1	1/53

(ii)



Giving initial authority value of 1 to all nodes.

a(n) = h(Y) = 1

a(A) = h(Y) + h(C) + h(D) + h(E) = 4.

a(B) = a(E) + a(F) = 2

a(G) = h(F) = 1

h(Y) = a(n) + a(A) = 5

h(C) = a(A) = 4

h(D) = h(A) = 4

h(E) = a(A) + a(B) = 6

h(F) = a(B) + a(G) = 3

hub total = 5+4+4+6+3 = 22

Authority total = 1+4+2+1 = 8

Normalized values:

$a(n) = 1/8$
$a(A) = 4/8 = 1/2$
$a(B) = 2/8 = 1/4$
$a(G) = 1/8$

$h(C) = 4/22$
$h(D) = 4/22 = 2/11$
$h(E) = 6/22 = 3/11$
$h(F) = 3/22 = 3/22$
$h(Y) = 5/22 = 5/22$

Iteration 2:

$a(n) = 5$
$a(A) = 5+4+4+6 = 19$
$a(B) = 6+3 = 9$
$a(G) = 3$

$h(C) = 19$
$h(D) = 19$
$h(E) = 19+9 = 28$
$h(F) = 9+3 = 12$
$h(Y) = 19+5 = 24$

hub total $= 19+19+28+12+24$
$= 102$

Authority total $= 5+19+9+3 = 36$

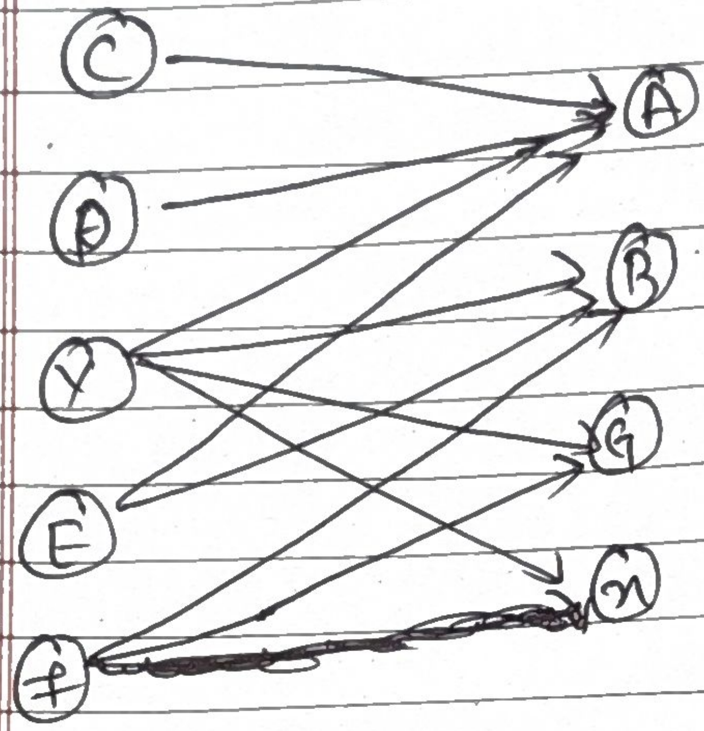
Normalized values:

$a(n) = 5/36$
$a(A) = 19/36$
$a(B) = 9/36 = 1/4$
$a(G) = 3/36 = 1/12$

$h(C) = 19/102$
$h(D) = 19/102$
$h(E) = 28/102 = 14/51$
$h(F) = 12/102 = 6/51$
$h(Y) = 24/102 = 12/51$

	Iteration 1 (Before)	(After)	Iteration 2 (Before)	(After)
$a(n)$	1	1/8	5	5/36
$h(Y)$	5	5/22	24	12/51

2.(c) (iii)



Initial Authorities for all node of 1.

$a(A) = h(C) + h(D) + h(Y) + h(E)$
$= 4$

$a(B) = h(Y) + h(E) + h(F)$
$= 3$

$a(G) = h(Y) + h(F) = 2$

$a(n) = h(Y) = 1.$

$h(C) = a(A) = 4$
$h(D) = a(A) = 4.$
$h(Y) = a(A) + a(B) + a(G) +$
$a(n) = 4+3+2+1 = 10$
$h(E) = a(A) + a(B) = 4+3$
$= 7.$
$h(F) = a(B) + a(G) = 3+2$
$= 5.$

hub-total $= 4+4+10+7+5 = 30$
Authority-total $= 1+2+3+4 = 10$

Normalized value:→

$a(A) = 4/10 = 2/5$	$h(C) = 4/30 = 2/15$
$a(B) = 3/10$	$h(D) = 4/30 = 2/15$
$a(G) = 2/10 = 1/5$	$h(Y) = 10/30 = 1/3$
$a(n) = 1/10$	$h(E) = 7/30$
	$h(F) = 5/30 = 1/6.$

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Iteration 2:

$a(A) = h(C) + h(D) + h(Y) + h(E) = 4 + 4 + 10 + 7 = 25$
$a(B) = h(Y) + h(E) + h(F) = 10 + 7 + 5 = 22$
$a(G) = h(Y) + h(F) = 10 + 5 = 15$
$a(M) = h(Y) = 10$

$h(C) = a(A) = 25$
$h(D) = a(A) = 25$
$h(E) = a(A) + a(B) = 25 + 22 = 47$
$h(Y) = a(A) + a(B) + a(G) + a(M) = 25 + 22 + 15 + 10 = 72$
$h(F) = a(B) + a(G) = 22 + 15 = 37$

hub-total $= 25 + 25 + 47 + 72 + 37 = 206$
Authority-total $= 25 + 22 + 15 + 10 = 72$

Normalized value:→
$a(A) = 25/72$ $h(C) = 25/206$
$a(B) = 22/72 = 11/36$ $h(D) = 25/206$
$a(G) = 15/72 = 5/24$ $h(E) = 47/206$
$a(M) = 10/72 = 5/36$ $h(Y) = 72/206 = 36/103$
$h(F) = 37/206$

	Iteration 1 (Before)	(After)	Iteration 2 (Before)	(After)
a(M)	1	1/10	10	5/36
h(Y)	10	1/3	72	36/103

Conclusion:→
- part (ii) & (iii) have the same & highest normalized authority score 5/36 for a(M).

Q.no.3. Beer sales Analysis

a. Two power law or rich get richer model:

The sale of beer mimics will follow a power law distribution, where a few brands account for most sales, and the rest hardly sell at all. This happens because the behavior of the customer is not at all random.

P customers choose their beer brand at random, without looking at the sales board, and the probability that any one model out of 5 beers will be chosen is 1/5; i.e., P(Molson) = P(Keith) = P(Stella) = P(Corona) = P(Heineken) = 1/5.

1. Preferential model:

- With probability **p**, the customer chooses uniformly at random (ignores the board):

$$P(\text{choose brand } i) = \frac{1}{5} \text{for } i = 1,2,3,4,5$$

- With probability **(1-p)**, the customer looks at the board and chooses a brand with probability proportional to its current sales count:

$$P(\text{choose brand } i) = \frac{S_i}{\sum_{j=1}^{5} S_j}$$

where S_i is the current number of bottles sold for brand i and S_j is the current sales count for brand j.

This is the classic "rich-get-richer" mechanism. If all brands start at 0, a small lead is established by the first few random customers in one brand's favor. Because of the preference for what most influenced customers bought, this lead further widens over time. The first brand to get that lucky break builds momentum and pulls away so much farther than it ends in a distribution that greatly skews in its favor over all other brands.

2. The copying Model:

The imitative model explains a portion of consumer behavior in which customers do not make choices independently or for themselves, but their choice-set is dependent on the choice of others, and thus perceived as imitations or copies.

First, Random Customers (P portion):

These customers don't survey the digital board. They just buy a brand of beer at random. All five brands have an equal chance of getting picked. They make independent decisions.

Second, Copying Customers (1-P portion):

The customers look at the digital board and see which brand has the highest sales. They then move on and buy the same brand that the previous customer bought. In fact, what they are doing is copying the earlier choices from other customers.

A copying customer walks into the store and looks at the digital board. Suppose that the brand that he sees has the most sales is the Corona. In this case, they reason to themselves saying, "Corona has the most sales, so it will be the best. That's why I will buy Corona too.

As a result, they buy Corona. The brand that gets an early lead keeps getting more customers simply because it is winning.

b. Morning initialization to boost high margin brand

The manager wants to shift the initialization of the digital board in early morning to place Heineken just right at the top, increasing sales of the highest profit margin, which in Heineken in this case.

The manager should initialize the board as (2, 3, 2, 5, 3) in the following order: Molson, Keith, Stella, Heineken, and Corona. Heineken is rated as a 5, so it should lead from the start. It will then trigger the copying model, where influenced consumers will see Heineken in the lead and consequently buy Heineken. The lead goes up all day, for more copying customers follow.

Values (2, 3, 2, 5, 3) are near enough to remain undetected and yet deliver a big edge to Heineken. Since Heineken is the highest brand margin, it will bring in huge sales and thus bring in huge profits for the store

C. suggestion for new Manager

The new manager can increase sales for the highest margin brand by using only honest marketing and merchandising strategies. The proposed brand of the highest margin should not be manipulated on the digital board. The legitimate techniques that can be executed include putting the high-margin brand at eye-level, creating visually appealing end-cap displays, using bright signage with 'Manager's Choice' or 'Customer Favorite,' offering slight discounts or loyalty bonuses, free sampling events, and positive customer reviews.

This increased attention and appeal gives the high margin brand an early sales lead, which then appears on the digital board. Once the brand shows as the leader, the same power law psychology from the copying model takes over naturally. The key advantage of this approach is that it achieves the same result as dishonest board manipulation making the high margin brand the market leader but through ethical, transparent, and sustainable means that build customer trust rather than damaging it.